

Harmonization review

We benchmark our model against existing harmonization methods, including **Harmony**, **Scaranoma**, and **scVI**. Additionally, we leverage **scETM** embeddings to extract meaningful information, aiming to reduce dataset size for downstream analysis while optimizing memory usage.

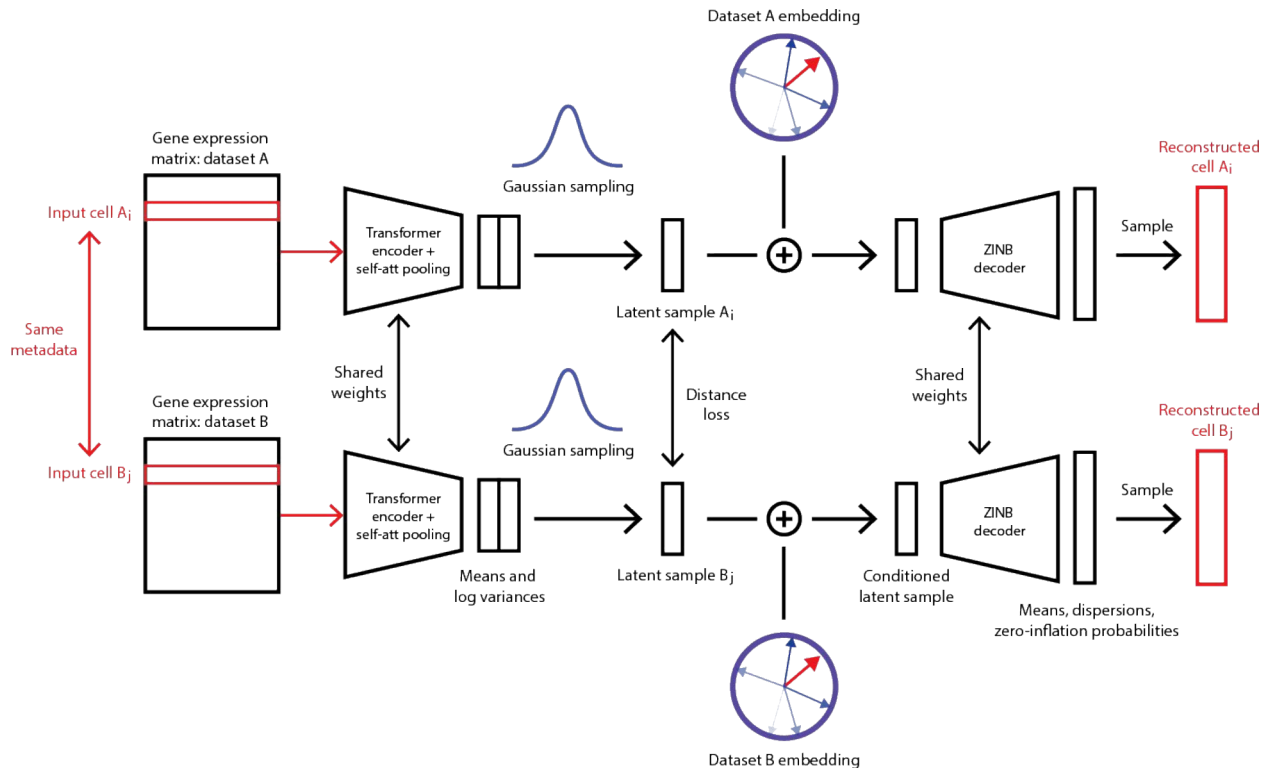
We benchmarked our model against established harmonization methods, including Harmony, Scaranoma, and scVI, using a common dataset.

- We generated **UMAP projections** to visually assess batch effect removal and verify whether cells from different batches are properly integrated. (Anti-PD1 datasets harmonized with parse)
- We performed **cell type classification** to ensure that harmonization does not alter the inherent biological identity of cells and that original cell types are preserved.
- We analyzed the model's ability to capture **perturbations** and identify **disease-specific signals** across different conditions, and further evaluated whether harmonized data from multiple datasets can accurately classify disease states while **preserving meaningful biological signals**. (perturbation dataset harmonized with parse)

For metric-based evaluation, we used kBET score, NMI, Graph iLISI, ASW (cell), and ASW (batch). For classification performance, we evaluated using F1-score and accuracy.

Architecture overview:

Variational Autoencoder with dataset embeddings - aligns latent representations of cells with same metadata.



Benchmarking against other integrations:

Initial findings from benchmarking each model against our proposed approach are summarized as follows:

Integration	kBET Score	NMI	Graph iLISI	ASW_cell	ASW_batch
Harmonized	0.234	0.551	5.544	0.285	-0.101
Harmony	0.033	0.490	4.028	0.328	-0.092
Scanorama	0.001	0.497	4.020	0.261	-0.055
PCA (Unharmonized)	0.001	0.501	3.989	0.316	-0.073
scVI	0.090	0.474	4.177	0.238	-0.028

Here are the UMAPS for these findings

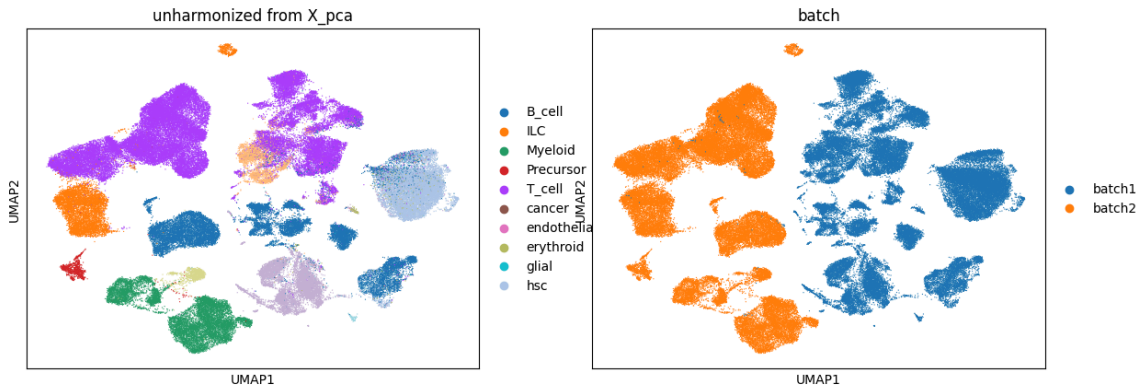


Figure 1: Unharmonized Data (PCA Representation)

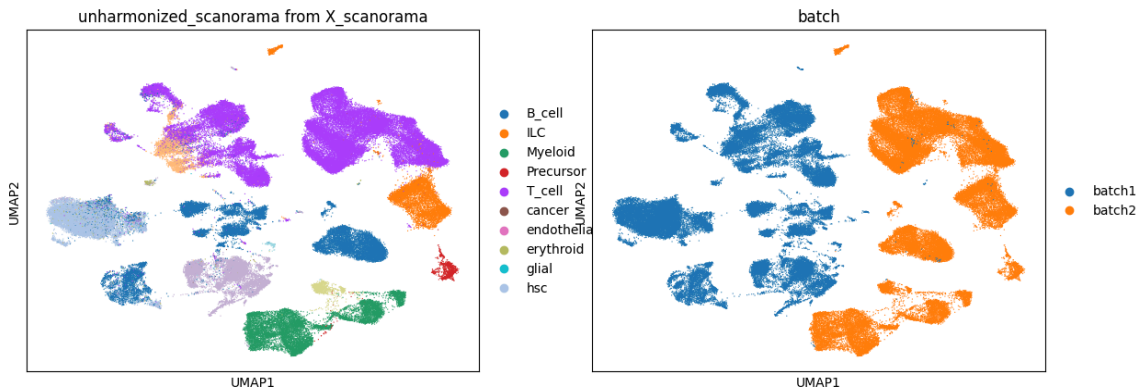


Figure 2: Scanorama Integration on Unharmonized Data

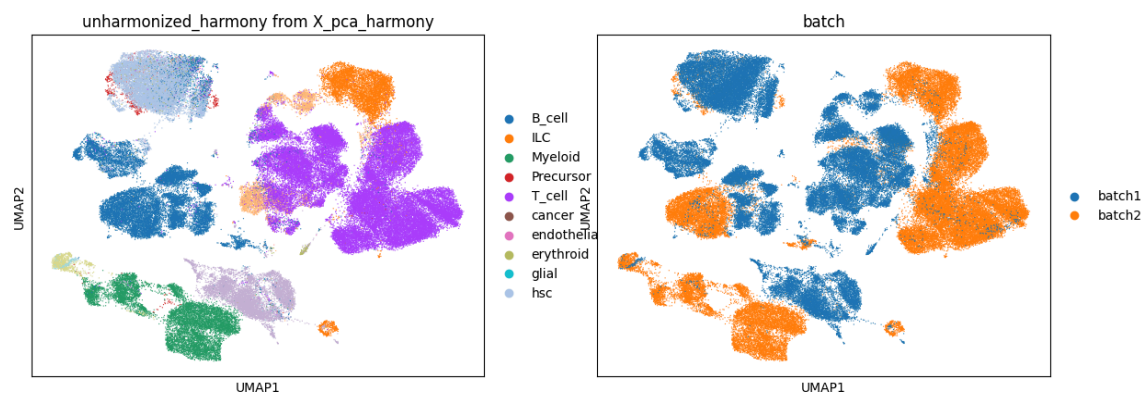


Figure 3: Harmony Integration on Unharmonized Data

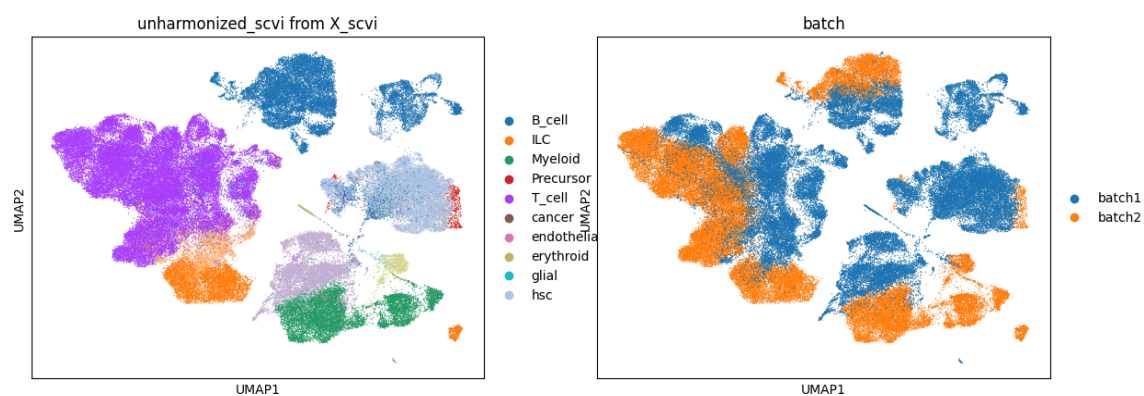


Figure 4: scVI Integration on Unharmonized Data

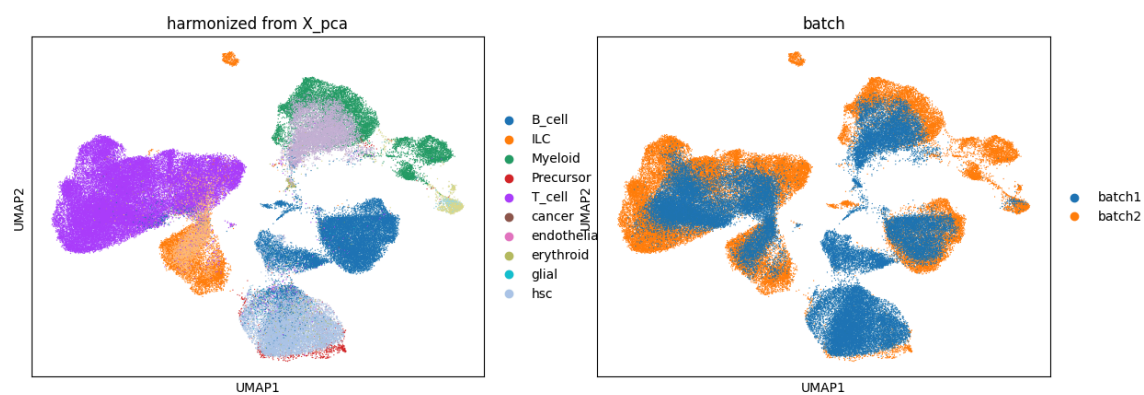
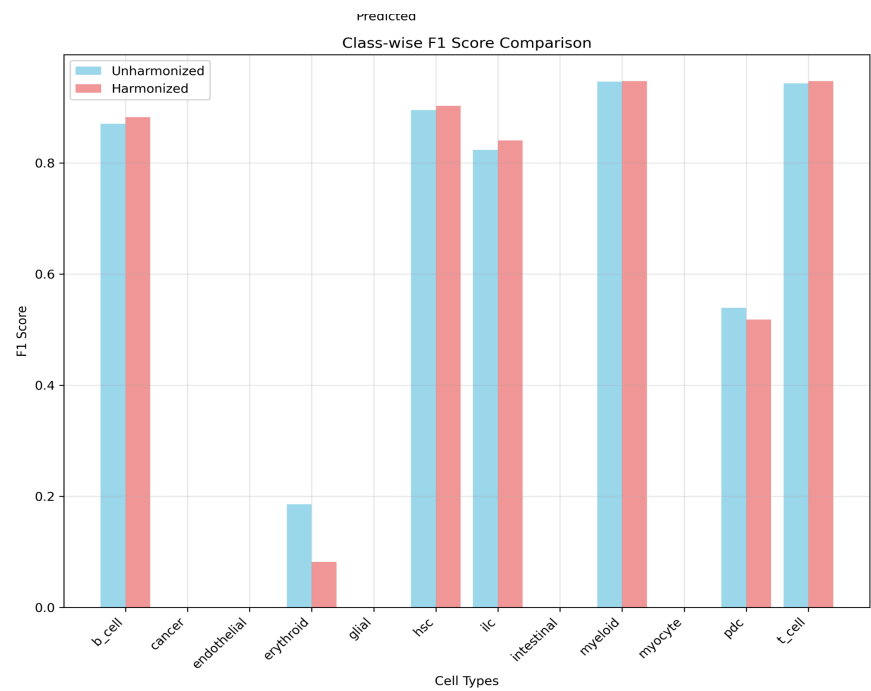
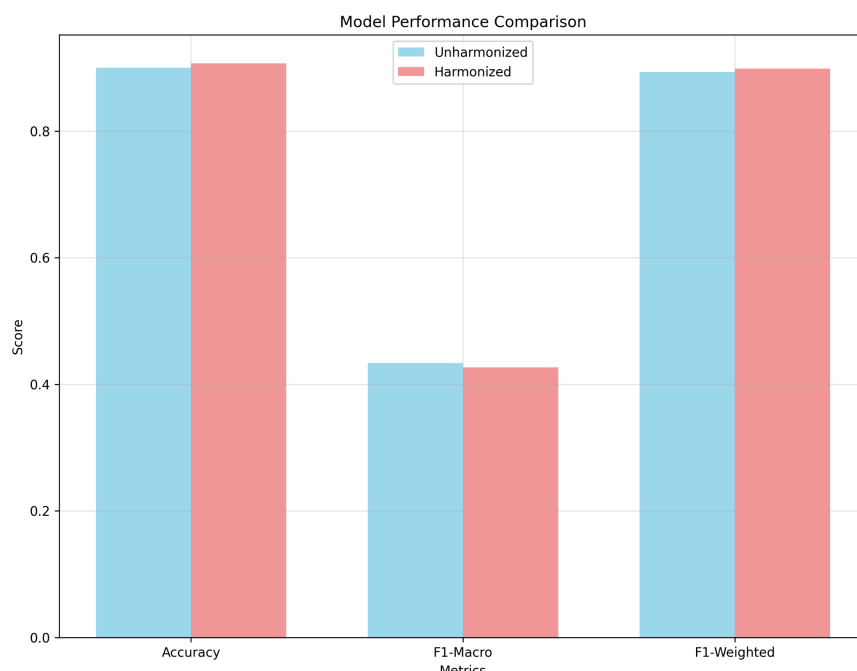


Figure 5: Harmonized Data (PCA Representation)

Classification of Cell-Types

For further evaluation of biological preservation, we adopted a classification-based approach. Initially, we performed cell type classification to assess whether the intrinsic cellular identities are retained after harmonization. For this analysis, we established a baseline by comparing classification performance between the unharmonized dataset and the dataset processed using our harmonization model.





Metric	Unharmonized	Harmonized
Accuracy	0.900095	0.906895
F1-Macro	0.433546	0.426643
F1-Weighted	0.893285	0.898646

In this setup, we were unable to observe a significant difference between the unharmonized and harmonized datasets, as their performance remained very similar. Consequently, we shifted our focus to perturbation analysis to better evaluate biological conservation.

Benchmarking against other model with scETM:

We further utilized scETM embeddings for feature extraction to assess whether meaningful biological information is preserved after data integration. scETM (single-cell Embedded Topic Model) captures latent gene expression patterns in a lower-dimensional space, enabling efficient representation while retaining key signals. The results are summarized as follows:

Integration (scETM Embeddings)	kBET Score	NMI	Graph iLISI	ASW_cell	ASW_batch
Harmonized	0.234	0.551	5.544	0.285	-0.101
Scanorama	0.030	0.486	4.653	0.388	-0.168
scVI	0.026	0.4345	5.526	0.221	-0.271
PCA (Unharmonized)	0.012	0.480	4.675	0.400	-0.175
Harmony	0.010	0.4809	4.698	0.413	-0.228

Here are the UMAPS for the results

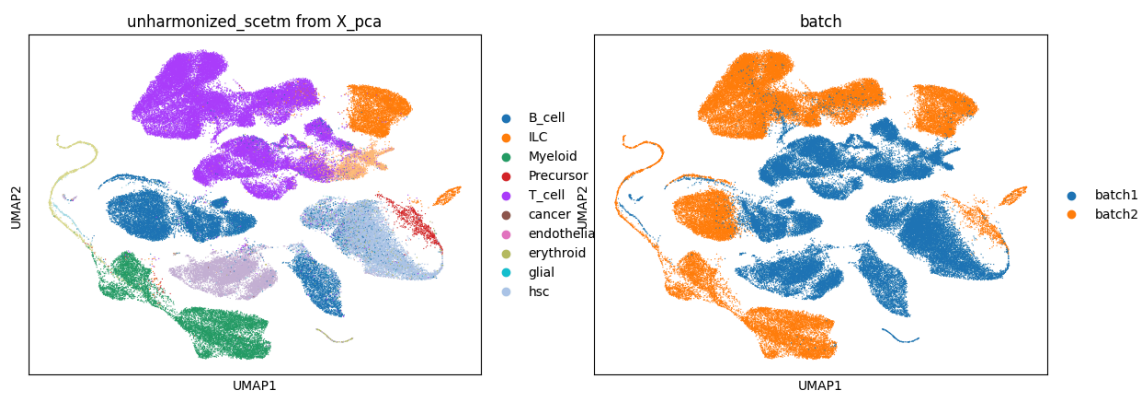


Figure 6: Unharmonized Data with scETM Preprocessing

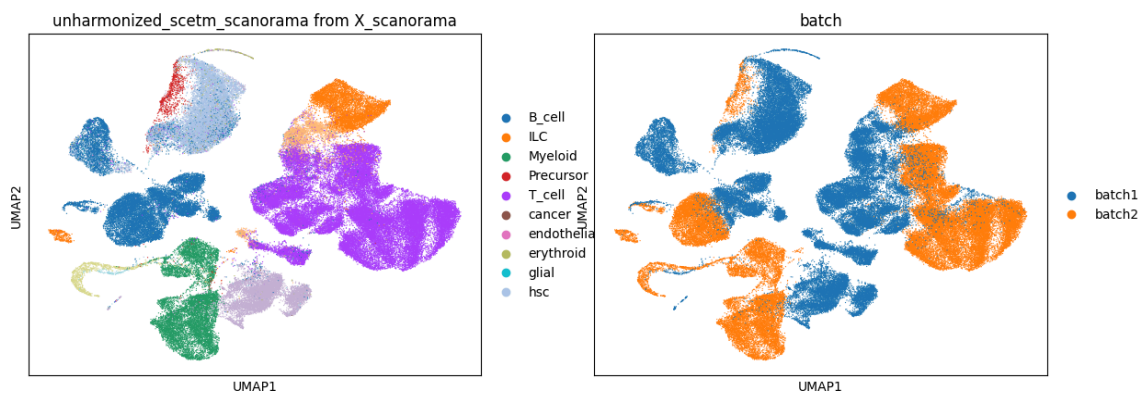


Figure 7: Scanorama Integration on scETM-Processed Data

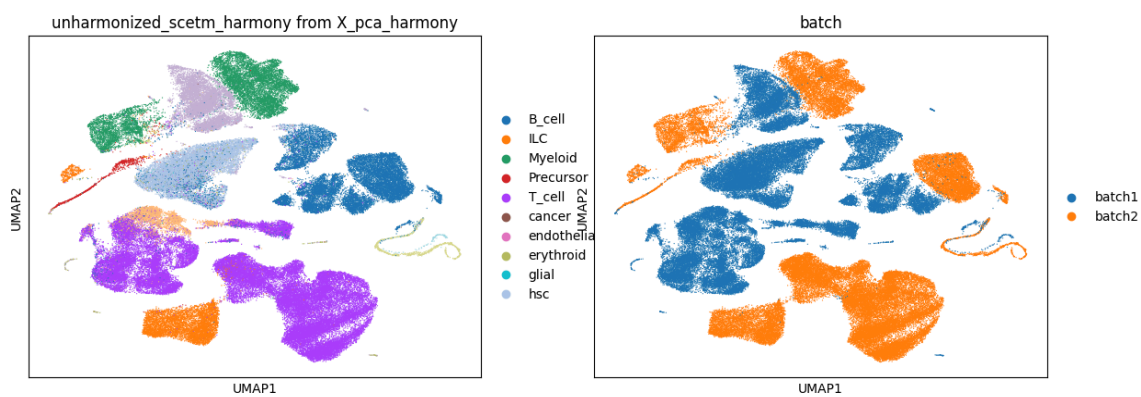


Figure 8: Harmony Integration on scETM-Processed Data

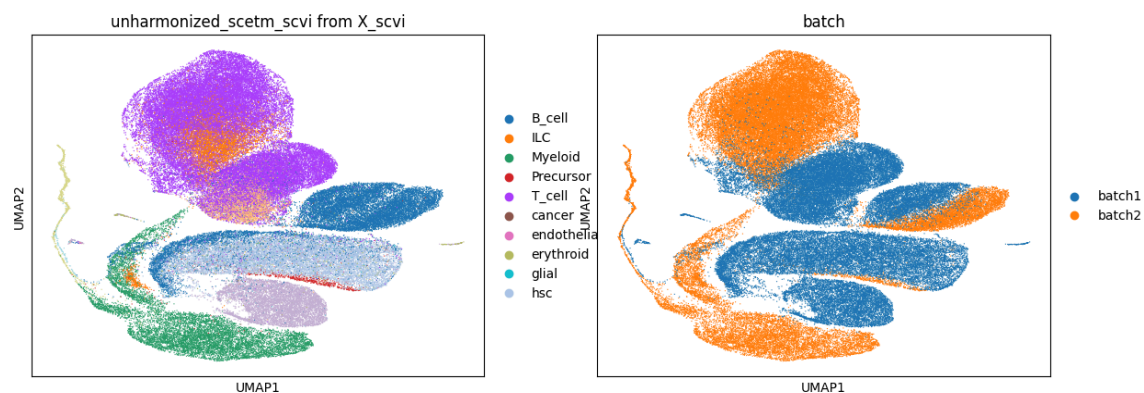


Figure 9: scVI Integration on scETM-Processed Data

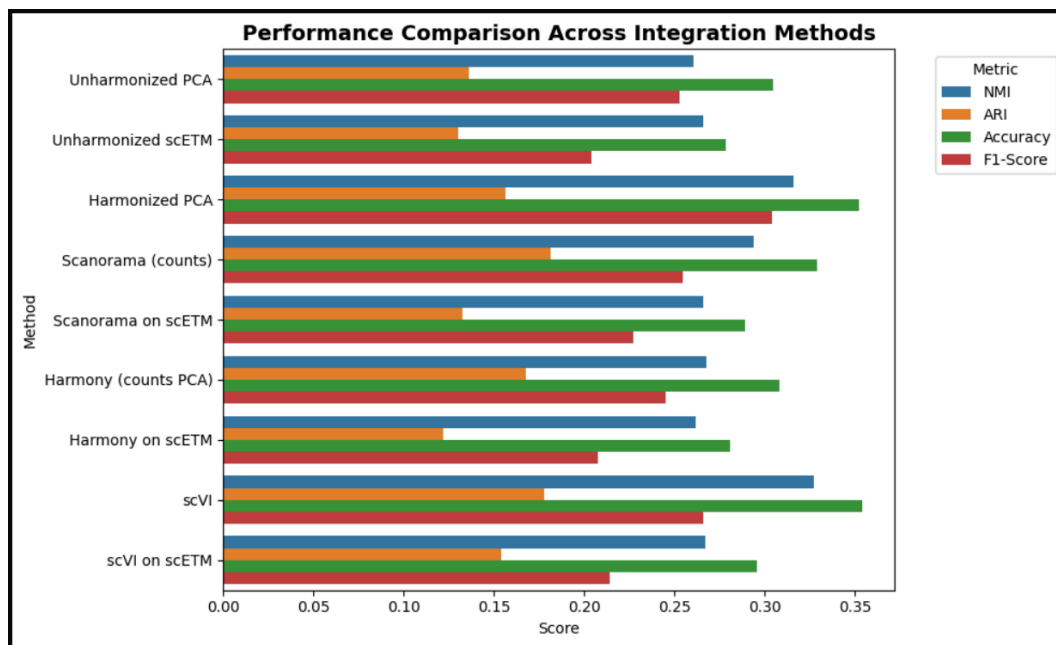
Perturbation Analysis :

The dataset, derived from a disease study with perturbation labels representing different conditions, was combined with Parse Biosciences data to introduce additional batch effects and increase integration complexity. Harmonization methods, including Harmony, Scanorama, scVI, and scETM-based approaches, were applied to correct these effects. KMeans clustering on the resulting embeddings was then compared with the original perturbation labels to evaluate how well each method preserved biological signals while removing technical variation.

Here are the results for all model and our model along with scTEM embeddings:

Method	NMI	ARI	Accuracy	F1-Score
Harmonized PCA	0.3162	0.1562	0.3523	0.3042
Unharmonized PCA	0.2605	0.1361	0.3045	0.2530
Unharmonized scETM	0.2663	0.1305	0.2787	0.2043

Scanorama (counts)	0.2939	0.1814	0.3292	0.2547
Scanorama on scETM	0.2662	0.1325	0.2891	0.2274
Harmony (counts PCA)	0.2676	0.1680	0.3084	0.2453
Harmony on scETM	0.2616	0.1219	0.2808	0.2079
scVI	0.3275	0.1776	0.3542	0.2663
scVI on scETM	0.2670	0.1542	0.2960	0.2141



Conclusions:

Bibliography: